



Farhad Shadmand

AI RESEARCHER AND DEVELOPER

CONTACT

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EDUCATION

PhD, Electrical Engineering & Intelligent Systems
University of Coimbra — Portugal
2021 – 2026

M.Sc., Complex Systems, Physics
Isfahan University of Technology
— Iran
2015 – 2017

Bachelor's, Physics
Isfahan University of Technology
— Iran
2011 – 2015

SKILLS

Programming
Python, C++, Java (Basic), HTML, CSS

AI / ML
PyTorch, TensorFlow, ONNX, PyTorch Lightning, Hugging Face, DeepSpeed, Scikit-learn, OpenVINO, XGBoost, LightGBM, CatBoost, torchdata, Accelerate, Keras, JAX, FLAX, Optax, Core ML

NLP
spaCy, NLTK, AllenNLP, Transformers, Tokenizers, Optimum (Intel / Neuron / Habana / ONNX), OpenAI SDK

AI Agents
LangChain, LangSmith, LangGraph, LangFlow, RAG, Gradio

Systems
Multi-GPU, CUDA, SLURM, Docker, Kubernetes, OpenHPC

CI/CD
Jenkins, GitHub Actions, GitLab CI

Computer Vision
OpenCV, dlib, Pillow

Data
SQL, MySQL, Spark, pandas, ChromaDB

LANGUAGES

Persian (Native)
English (Fluent)
Portuguese (Beginner)

Researcher and developer with hands-on experience building deep learning models based on diffusion models, GANs, and VLM-based systems on multi-GPU platforms (CUDA). Work has led to multiple publications, including CVPR, WACV, and IEEE Access — with strong skills in PyTorch, computer vision, model optimization, and AI agents.

EXPERIENCE

Post-doctoral Researcher 2026 – Present
VisTeam, Institute of Systems and Robotics – Coimbra (ISR) Coimbra, Portugal

- Research on generative video models, including video watermarking and diffusion-based generation (Stable Diffusion).

Stack: PyTorch, Hugging Face, Accelerate, Slurm, OpenCV, Weights & Biases (wandb)

Research Intern Aug 2025 – Dec 2025
Adobe Paris, France

- Developed robust watermarking models using geometric symmetry-aware convolutional networks combined with a VLM-based generative noise-simulation pipeline to accurately replicate print-and-scan degradation effects.
- Significantly enhanced watermark resilience against real-world distortions introduced during printing, scanning, and image recapture.

Stack: Deep Learning, AWS, PyTorch Lightning, data distribution

Manager: Dr. Shruti Agarwal (shrugarw@adobe.com)

Researcher Sep 2019 – Jul 2025
VisTeam, Institute of Systems and Robotics – Coimbra (ISR) Coimbra, Portugal

- Developed deep learning models for steganography, watermarking, object detection, face detection, and face verification.
- Contributed to the projects VISUAL-ID, TruIM, and FACING, resulting in multiple publications and practical solutions for identity-document security.

Stack: Deep learning & computer vision libraries (OpenCV, dlib, Pillow), C++, Django, FastAPI

Researcher 2018 – 2019
Instituto Universitário de Lisboa (ISCTE) Lisbon, Portugal

- Began research at ISCTE and the Faculty of Sciences of the University of Lisbon (FCUL), under the supervision of Prof. José Carlos Dias.
- Applied machine learning, deep learning, and NLP to analyze financial markets, processing data from news sites, Twitter, and stock-market platforms to build predictive models for financial trends.

Stack: NLP libraries, Python, MATLAB, Requests, BeautifulSoup, REST API

PATENTS

Encoding, Decoding and Integrity Validation Systems for a Security Document with a Steganography-Encoded Image

Granted Oct. 2025 · Inventors: Nuno Gonçalves and Farhad Shadmand · Applicants: Imprensa Nacional Casa da Moeda and University of Coimbra

Encoding, decoding and integrity validation systems for a security document with a steganography-encoded image, and associated methods, computer programs, and computer-readable data carrier. PT117136A (Portugal) · EP4064095A1 (European Patent Office)

PUBLICATIONS

StampOne: Addressing Frequency Balance in Printer-proof Steganography
CVPR (2024)

CodeFace: A Deep Learning Printer-Proof Steganography for Face Portraits
IEEE Access 9 (2021)

StylePuncher: Encoding a Hidden QR Code into Images
ICPRAM (2025)

DocSafe: Towards Practical Print-Proof Image Steganography via Frequency Decomposition and Covariance Alignment
IEEE Access (2026.3680290), 2026

ENTERTAINMENTS

Dance (Afro-Latin, Forro)
Swimming
Movie

RiemStega: Covariance-based Loss for Print-Proof Transmission of Data in Images
WACV (2025)

Young Labeled Faces in the Wild (YLFW): A Dataset for Children Faces Recognition
IEEE 18th Int'l Conf. on Automatic Face and Gesture Recognition (FG) (2024)

MorDeephy: Face Morphing Detection via Fused Classification
12th Int'l Conf. on Pattern Recognition Application and Methods (2022)

Towards Facial Biometrics for ID Document Validation in Mobile Devices
Applied Sciences 11.13 (2021)